

1. How Loud are Sounds?

In the previous note, we saw that atmospheric pressure is close to 100,000 Pascals, equivalent to about 1 kg per square centimetre. It may not surprise you to hear that what we hear as sound comes from changing pressure, so when we talk about how “loud” a sound is, what we really mean from a physics perspective is “how much is the pressure changing?” It’s probably not a lot, at least relative to atmospheric pressure, because sounds don’t knock us over while a good gust of wind would. How much lower, though?

We’ll try to make a reasonable estimate based on some sensible guesses. The answer won’t be perfect, but we’ll at least be in a position to say “is it a thousandth of atmospheric pressure? A millionth? A billionth?” This sort of answer is called an “order-of-magnitude estimate” – getting a perfect answer would be much, much harder, but at least we can get started with our order-of-magnitude guess. Instruments like flutes or violins or trombones would be hard to estimate, because it’s really difficult to know how much of the work a musician is doing ends up as sound. In fact, the answer is usually “not very much” because most of the energy ends up getting dissipated as heat in the instrument rather than coming out as sound. One case we can estimate is a piano. Because of the way a piano is played, we can put an upper limit on how much energy goes into sound. Presumably nearly all of you have heard a piano in a concert at some point as well.

1.1. How Much Work in a Piano Note?

The most basic definition of work is “force times distance.” If I know how much force I exerted on something, and how far I pushed it, I know how much work I did. The metric unit of work or energy is a Joule, the work required to apply a force of one Newton over one metre. As an example, how much work does it take to climb a flight of stairs? Let’s say you’re 100 kg, and a flight of stairs goes up 3m. Newton’s second law is $F = ma$, force equals mass times acceleration, and at the Earth’s surface, the acceleration from gravity is about 10 m/s^2 . The force from gravity on us is then $100\text{kg} \times 10\text{m/s}^2 = 1000\text{kgm/s}^2 = 1000\text{N}$. To climb the stairs, we have to exert a force of at least 1000 N upwards to counteract gravity, and that force is doing the work. If we climb 3m, then force times distance is $1000\text{N} \times 3\text{m} = 3000\text{N} \cdot \text{m} = 3000\text{J} = 3\text{kJ}$. In most of the world, the energy in food is listed directly in kJ , but here in Canada, it’s usually in kilo-calories, which we usually just call calories. One (kilo-)calorie is 4.19 kJ , so it takes a bit less than a calorie to climb a flight of stairs.

Let’s redo the same calculation, except now for a piano. We don’t know how much of the work we do will end up as sound, but we know it can’t be more than the work we did pressing a key. How much work do we do pressing a key? Work is force times distance, so let’s start with a distance. Picture a piano in your head - about how far does a key go down? It’s a lot more than a millimetre, but a lot less than a metre, so 1 cm seems reasonable. Your intrepid instructor measured the key travel on the piano at his house, and indeed the travel is almost exactly 1 cm. How about force? One way to estimate is to imagine different weights on a piano key. What sort of weight would it take resting on a key to push the key down and make a sound? Well, your instructor set a bottle of

whiskey on the keyboard and let it push some keys down. The bottle weight 1.3 kg, and it pushed down 4 keys at what sounded like a *mezzo-piano* to my ear. The total force is $1.3 * 10 = 13N$ split amongst 4 keys, so the force on each key was $13/4 \sim 3N$. A force of $3N$ over a distance of $1cm = 0.01m$ then is a total work of $0.03J$. The amount of energy might be (much) less than this, but it cannot be more.

Since we can hear a pianist playing at a *mezzo-piano* volume inside a concert hall, we know that $0.03J$ will be easily heard. How do we convert that work to a change in pressure, though? For simplicity, we can assume we have a partition in our concert hall with half the air on one side and half on the other. We push on that partition, compressing the air on one side and expanding it on the other, until we've done our $0.03J$ of work. You're hopefully now getting an intuitive sense for just how small the pressure changes from typical sounds are. A thirtieth of a Joule, spread across a concert hall, is not going to get you a very large pressure change.

Look at a picture of Montreal's Maison Symphonique. The stage is about 20m across, the hall is about as high as it is wide, and it's a few times longer than it is wide. For ease, let's call the dimensions 20x20x50m. I set up my partition that splits the hall in half, so now I have two volumes of 10x20x50, and I'm going to move along the 10m direction. What is the net force on the partition? Well, if I haven't moved the partition yet, the net force is zero because the pressures on both sides match. What happens after I move it a distance dx though? Well, now the density on one side has gone up by a factor of $dx/10$, and the density on the other side has gone down by the same amount. For simplicity, we'll assume the temperature stays the same¹. From the ideal gas law with temperature fixed, we know pressure times volume is a constant, so whatever factor the volume went down is, the pressure must have gone up by the same factor. Let's say we move our partition by 1 cm, then what happens? The new volume is $(10m-0.01m)/(10m) = (1-0.001)m$, or one part in a thousand smaller. That means the new pressure is a part in a thousand larger, so the change in pressure is $100 \text{ kPa}/1000=100 \text{ Pascals}$. Force is pressure times area, so the force on our partition 100 times the area of the partition. We're moving in the 10m direction, so the area is 20m by 50m, or $1000m^2$. The total force on the partition is $100*1000$ or $100,000N$. The work we did was that force times the 1cm distance we moved, so $100,000N \times 0.01m = 1000J$. That's way, way more than the $0.03J$ we have, so the distance moved must be much less than 1 cm. At this point, the easiest way to get our actual answer is to take what happens at 1 cm, and adjust it to get the answer we want. If we move by a distance dx , then the pressure changes by a factor of dx/x , so the force must be proportional to dx/x as well. The work is force times distance, and the distance is dx , so the work must scale like dx^2/x . Since x doesn't change (after all, the room size is fixed), so the work we do goes like dx^2 . The actual work we did was $0.03J$, so we must have

$$(dx/0.01)^2 = (0.03/1000) \tag{1}$$

$$dx/0.01 = \sqrt{0.03/1000} \tag{2}$$

$$dx = 0.01 \times \sqrt{0.03/1000} = 5.47 \times 10^{-5}m \sim 50\mu m \tag{3}$$

Now we have at least a rough estimate for sound pressure. Our piano note in a concert hall

¹In fact, this is not a good assumption as the air on the compressed side heats up as well, but we can ignore that for our order-of-magnitude estimate.

moves the partition by about 50 microns out of 10 metres, a fractional change of 5×10^{-6} . A fractional volume change of five parts per million will lead to a similar fractional pressure change, and since the pressure is 100,000 Pascals, the *change* in pressure is on the order of half a Pascal. We know this is an overestimate, because we assumed all of the energy from pressing the piano key ended up as sound, and that none of that sound leaked out of the concert hall, but at least we're in the ballpark. If you go to *e.g.* Wikipedia, you'll see that sound pressures of half a Pascal is actually really loud - about 88 dB, or just above the threshold where hearing protection is required to prevent hearing loss. The true answer is about a factor of 10 or so lower, or an order-of-magnitude. We call these order-of-magnitude estimates for a reason!

For fun, let's see if we can do even better. There are two main assumptions we made that probably weren't very good. The first is that all the energy in the key press ended up as sound. It's tricky to estimate this well - to get an idea, there's a paper by Seok Jae Bang (2009) that estimates the acoustic efficiency of a Yamaha upright. They find that the efficiency is only about 1%, so we should decrease our sound energy by a factor of 100. Second, we assumed that all the sound energy ended up in the hall *at the same time*. That's also obviously wrong. Why? Well, if I play a piano note and hold the key down, it takes quite a while (many seconds) for the sound volume to drop, so the energy takes a while to go from vibrations of the piano strings to vibrations of the air, say 10 seconds. However, in a concert hall, if I make a noise, the time it takes the sound volume to drop (the reverb time) is a lot less, more like a second. That means that at any given time, only about 1s (reverb)/10s (piano sustain) or 10% of the total sound energy we'll put out is contained in the hall. With these correction factors, our sound energy is down by a factor of 1000 from our initial overestimate, so the pressure is down by a factor of the square root of that, so a better estimate is 0.01 Pascal for our sound pressure. That's actually pretty good - Wikipedia tells us that's in the range of normal conversation (0.002-0.02 Pascals, or 40-60 dB).

One final calculation, also for fun, I'd like to do is to compare the sound pressure we just calculated to change in pressure from changing altitude. The height scale of the atmosphere is close to 10 km, which is why when you climb a tall mountain you notice there's a lot less air, but not when you climb stairs. If we assume the pressure goes from 100 kPa to 0 when we go up 10 km (again, we're after a rough estimate), then the change per metre is $100,000/(10\text{km}=10,000\text{m})$, or 10 Pascals/metre. Just standing up will change the pressure by 10 Pascals, which is much, much bigger than the sounds! Our 0.01 Pascal improved estimate for a piano volume corresponds to $0.01/10=0.001\text{m}$ or 1 mm. Because we (usually) change altitude slowly and smoothly, our ears do not register those pressure changes as sounds. If they did, we'd all go deaf just from getting out of bed in the morning.